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Hardware

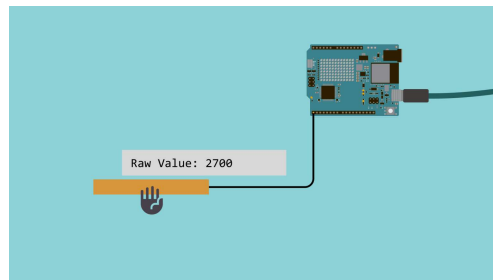
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UNO R4 Capacitive-Touch Tutorial

Learn to use the built-in capacitive sensing capabilities of the Arduino® UNO R4.

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Capacitive sensing is a technology that detects changes in capacitance to determine the presence or absence of a conductive object, such as a human finger. This principle is widely used in touch-sensitive devices. The Arduino® UNO R4, both the **WiFi** and **Minima** versions, come equipped with built-in capacitive sensing capabilities, making it easier to integrate touch inputs into your projects.



Required Hardware

To use the library, you will need one of the compatible boards:

[Arduino UNO R4 WiFi](#)[Arduino UNO R4 Minima](#)

You will also need:

Wire

Something that can affect the electric field to touch (metal objects, materials with moisture, even bananas! Experimentation is key!)

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Required Hardware

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How Capacitive Sensing Works

Imagine the sensor as creating an invisible electric field around itself. When a conductive object (like your finger) approaches, it's like dropping a stone into a calm pond - the field gets disturbed. The sensor detects this disturbance as a change in capacitance, which is essentially how much electrical charge the system can store. Here's the process:

- 1 The sensor electrode creates an electric field by applying a small voltage
- 2 When you approach or touch the sensor, your body becomes part of the capacitive circuit
- 3 This increases the capacitance between the sensor electrode and ground
- 4 The microcontroller detects this capacitance change (often by measuring timing differences) and determines if it's significant enough to register as a "touch"

Using Capacitive Sensing on the UNO R4

The UNO R4 features a Capacitive Touch Sensing Unit (CTSU) that allows you to use certain pins as capacitive touch inputs. To utilize these capabilities, you can use the [Arduino_CapacitiveTouch library](#).

Compatible Pins

For both the UNO R4 WiFi and Minima boards, the compatible pins for capacitive touch are listed in the [Arduino_CapacitiveTouch library documentation](#).

Arduino UNO R4 Minima:

Arduino Pin	Touch Sensor Channel (TS#)	Channel Control Index (CHAC idx)	Channel Control Bit Mask (CHAC val)
D0	9	1	(1 << 1)
D1	8	1	(1 << 0)
D2	34	4	(1 << 2)
D3	13	1	(1 << 5)
D8	11	1	(1 << 3)
D9	2	0	(1 << 2)
D11	10	1	(1 << 2)
D13	12	1	(1 << 4)
A1 (D15)	21	2	(1 << 5)
A2 (D16)	22	2	(1 << 6)
LOVE_BUTTON	0	0	(1 << 0)

Arduino UNO R4 WiFi:

Arduino Pin	Touch Sensor Channel (TS#)	Channel Control Index (CHAC idx)	Channel Control Bit Mask (CHAC val)
D0	9	1	(1 << 1)
D1	8	1	(1 << 0)
D2	13	1	(1 << 5)
D3	34	4	(1 << 2)
D6	12	1	(1 << 4)
D8	11	1	(1 << 3)
D9	2	0	(1 << 2)
D11	7	0	(1 << 7)
D12	6	0	(1 << 6)
A1 (D15)	21	2	(1 << 5)
A2 (D16)	22	2	(1 << 6)
LOVE_BUTTON	27	3	(1 << 3)

The **Arduino_CapacitiveTouch** library provides several functions to work with capacitive touch inputs:

`CapacitiveTouch(uint8_t pin)`

Constructs a capacitive touch sensor for the given pin.

`bool begin()` Initializes the sensor and configures the pin and hardware.

`int read()` Reads the raw sensor value and returns it.

`bool isTouched()` Checks if the sensor is touched based on the threshold.

`void setThreshold(int threshold)`

Sets the detection threshold for touch sensitivity.

`int getThreshold()` Retrieves the

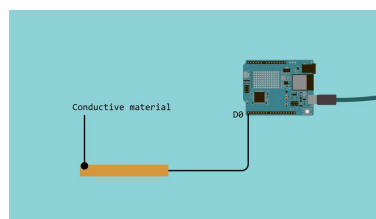
current detection threshold.

For more detailed usage and examples, refer to the [CapacitiveSensor library documentation](#).

Detecting Touch Example

Circuit

Here's a simple example to get you started with capacitive sensing on the UNO R4. For this example, we are connecting a single piece of any conductive material to the pin `D0` on the Board.



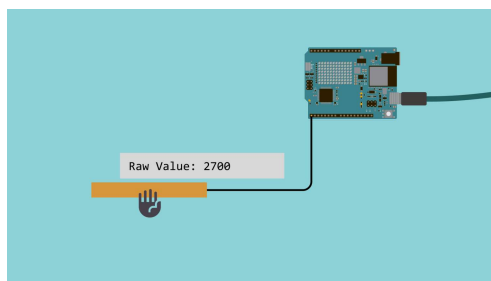
How to connect

Code

Note: To install the

`Arduino_CapacitiveTouch` library, open the Arduino IDE Library Manager, search for

```
1 #include "Arduino_CapacitiveTouch.h"
2
3 CapacitiveTouch touchButton = CapacitiveTouch(5, 10);
4
5 void setup() {
6   Serial.begin(9600);
7
8   if(touchButton.begin()){
9     Serial.println("Capacitive Touch Button initialized");
10  } else {
11    Serial.println("Failed to initialize Capacitive Touch Button");
12    while(true);
13  }
14
15  touchButton.setThreshold(200);
16 }
17
18 void loop() {
19   int sensorValue = touchButton.getRawValue();
20   Serial.print("Raw value: ")
21   Serial.println(sensorValue);
22
23   if (touchButton.isTouched())
24     Serial.println("Button touched");
25 }
26
27 delay(100);
28 }
```



Creative Project Ideas

Now that you have learned the basics, here are some fun and interesting project ideas:

Touch-Controlled Night Light - Create a lamp that turns on/off with a touch

Capacitive Touch Piano - Use multiple pins to create touch-sensitive keys

Smart Home Controller - Touch different areas to control various devices

Conclusion

Capacitive sensing on the Arduino UNO R4 lets you add intuitive touch controls with minimal hardware. Simply pair the board with the `Arduino_CapacitiveTouch` library. Tune the detection threshold for your environment (such as humidity and nearby electronics), and you can quickly scale from a single-touch button to richer, multi-point interfaces.

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